REQUIREMENT ANALYSIS — ARTIFACT 07

Technology Stack Architecture

Enterprise Frameworks, Cloud Infrastructure & Hardware Integration

1. LAYERED TECHNOLOGY BREAKDOWN

Frontend & Mobile Applications

React Native (iOS & Android)
Guest Mobile Application with native Bluetooth Low Energy (BLE) bindings for digital keyless door unlocking.

React.js & Tailwind CSS
Responsive Web Dashboard for Front Desk staff, Concierge, and Executive Analytics.

Backend Microservices & APIs

Node.js (TypeScript) & NestJS
Scalable microservices for real-time booking management, room assignment, and notification engines.

GraphQL & REST Gateway
Unified API access layer connecting internal staff modules and third-party POS endpoints.

Database & Caching Layer

PostgreSQL (AWS Aurora)
Relational DB with multi-AZ replication for guest records, billing transactions, and room inventories.

Redis Cluster
In-memory caching for session management, dynamic availability checks, and real-time door lock tokens.

Cloud & Hardware Integrations

AWS Cloud Infrastructure
Docker containers managed via AWS EKS (Kubernetes) with automated auto-scaling groups.

IoT Gateway & MQTT Protocol
Real-time lock communication and energy management sensor streaming across resort villas.

2. TECH STACK ECOSYSTEM DIAGRAM

